

Tianchu Zeng

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PROFESSIONAL SUMMARY

NUS PhD candidate in computational neuroscience and AI for science, developing methods for scalable brain modeling and statistically valid machine-learning evaluation. Developed DELSSOME, a deep-learning framework that accelerates biophysical brain-model fitting by 50–100×, enabling lifespan mapping of model-derived cortical excitation–inhibition (E/I) ratio in 12,005 participants. Co-developed SHARP, a statistical method for valid model comparisons under cross-validation, and co-led a meta-analysis of biomedical AI studies. Independently developed an AI-agent workflow for evidence extraction and statistical re-analysis. Co-first author of a study accepted for publication in Nature Methods.

RESEARCH INTERESTS

Computational neuroscience · AI for science · statistical evaluation of machine learning · brain foundation models

EDUCATION

National University of Singapore — Singapore

Aug 2022 - Present (expected 2027)

Ph.D. in Electrical & Computer Engineering · GPA: 4.88/5.0

Advisors: Assoc. Prof. BT Thomas Yeo and Prof. Thomas E. Nichols (University of Oxford)

Tsinghua University — Beijing, China

Aug 2018 - Jun 2022

B.Eng. in Department of Automation · GPA: 3.74 / 4.00

RESEARCH PORTFOLIO

DELSSOME — Scalable brain modeling and lifespan E/I trajectories

Aug 2022 - Present

Co-first-author study · Nature Methods, accepted

- Developed DELSSOME, a deep-learning framework to predict how well candidate brain-model parameters match empirical connectivity data. Accelerated fitting by 50–100× across three brain circuit model families while preserving fitting accuracy.
- Co-led modeling of 12,005 participants across 14 datasets to map normative lifespan trajectories of model-derived cortical E/I ratio. Identified a decline through development and adulthood, a late-life increase, sex differences, and a persistent sensory–association gradient.

SHARP — Valid model comparisons under cross-validation

2024 - Present

Co-first-author preprint · Under review

- Co-developed SHARP to test whether model performance differences are statistically supported, redesigning cross-validation to estimate variance and correlation. Across 420 simulation scenarios, it achieved the best balance of false-positive control and statistical power among 13 tests.
- Co-led a PRISMA-guided meta-analysis of 184 biomedical AI studies across 30 fields; 97% used tests that ignored between-fold correlation. Among re-analyzed studies whose abstract claims relied on invalid tests, 59% had at least one supporting comparison lose significance after accounting for this correlation.
- Developed an AI-agent workflow based on SHARP's meta-analysis procedures, supporting evidence extraction and reproducible statistical re-analysis of published model comparisons.

Brain Foundation Models — Rigorous benchmarking for phenotypic prediction

2025 - Present

- Build and fine-tune brain foundation models; benchmark phenotypic prediction across foundation models and against classical machine-learning methods.

SELECTED PUBLICATIONS

* Co-first author; + co-corresponding author; my name in bold. Author lists abbreviated.

1. **Zeng, T.***, Tian, F.*, Zhang, S., ..., Yeo, B. T. T., & Alzheimer's Disease Neuroimaging Initiative (2026). Optimizing biophysical large-scale brain circuit models with deep neural networks. *Nature Methods*. Accepted for publication. [Preprint](#).
2. **Zeng, T.**, & Yeo, B. T. T. (2026). Deep learning charts the excitation-inhibition balance of the brain throughout the lifespan. *Nature Methods*. Invited Research Briefing. Accepted.
3. **Zeng, T.***, Li, H.*, Zhang, S.*, ..., Nichols, T. E.+ & Yeo, B. T. T.+ (2026). Spurious model comparisons are widespread in biomedical artificial intelligence. *bioRxiv* preprint. Under review. [DOI](#).
4. Zhang, S.*, Roemer-Cassiano, S. N.*, Chong, J. R.*, ..., **Zeng, T.**, ..., Alzheimer's Disease Neuroimaging Initiative, Lai, M. K. P.+ & Yeo, B. T. T.+ (2026). Excitation-Inhibition Imbalance Links Amyloid Pathophysiology to Cognition in Non-Demented Individuals. *Alzheimer's & Dementia*. Accepted for publication.
5. Qu, Y. L., ..., **Zeng, T.**, ..., & Holmes, A. J. (2025). Distinct brain network features predict internalizing and externalizing traits in children, adolescents and adults. *Nature Mental Health*, 3(3), 306-317.
6. Zhang, S.*, Larsen, B.*, Sydnor, V. J.*, **Zeng, T.**, ..., Satterthwaite, T. D.+ & Yeo, B. T. T.+ (2024). In vivo whole-cortex marker of excitation-inhibition ratio indexes cortical maturation and cognitive ability in youth. *PNAS*, 121(23), e2318641121.

SCIENTIFIC SOFTWARE AND OPEN SOURCE

[DELSSOME](#) — Lead Developer

- Lead development and maintenance of the official implementation for the Nature Methods study, including group- and individual-level pipelines, pretrained models, reproducible examples, documentation, and regression tests.
- Built a workflow that fits biophysical models to functional connectivity and functional connectivity dynamics. It estimates model-derived cortical E/I ratios and validates fitted parameters through numerical simulation.

[SHARP](#) — Lead Developer

- Developed and publicly released the SHARP Python package for statistical model comparison under cross-validation.

SHARP Meta-analysis Agent Workflow — Sole Developer

- Implemented source-referenced reporting, automated checks of evidence references and statistical results, and selective second-agent review; preparing public release.

[ThomasYeoLab CBIG](#) — Research Software Contributor

- Contribute code, pull-request reviews, and maintenance to CBIG, an open-source neuroimaging software repository.

[admissions-ops](#) — Creator and Maintainer

- Built a local-first AI-agent workspace for graduate-application planning, with source-backed reports, automated validation, and human-review checkpoints.

[career-ops](#) — Contributor

- Contribute to the career-ops open-source project.

ADDITIONAL RESEARCH

Brain Parcellation with Resting-State fMRI — National University of Singapore

Aug 2022 - Jun 2023

- Developed machine-learning methods for group- and individual-level parcellation using functional-connectivity analysis and cortical-surface visualization.

Multi-Agent Traffic Trajectory Prediction — Tsinghua University

Jan 2020 - Jun 2022

- Built RNN-, GNN-, CVAE-, and Transformer-based traffic trajectory predictors; the work produced two co-first-authored peer-reviewed conference papers and one granted Chinese patent.

INDUSTRY EXPERIENCE

R&D Engineer Intern — Beijing Zhangshangxianji Network Technology Co., Ltd.

Jun 2021 - Sep 2021

- Implemented Spring Boot transaction services and unified Single Sign-On; built SMS authentication, CAPTCHA verification, and department-wide messaging and automated-call alerts.

SELECTED HONORS AND ACTIVITIES

- Best Poster Award (Basic Science), NUHS Scientific and Innovation Summit, 2026.
- NUS Research Scholarship, 2022-2026.
- Invited talk at the Asia-Pacific Computational and Cognitive Neuroscience Conference, RIKEN, 2025.
- Posters at OHBM 2026, NUHS 2026, and AI4X 2025.
- Teaching Assistant for EE2026 Digital Design and EE3731C Signal Processing at NUS.

TECHNICAL SKILLS

- **Machine learning and scientific computing:** Python, MATLAB, PyTorch, JAX, scikit-learn, SciPy; deep learning; LLM agents
- **Statistics: statistical inference for** cross-validation, hypothesis testing, simulation-based validation, false-positive control, power evaluation, confidence-interval calibration
- **Neuroimaging:** resting-state fMRI, functional connectivity, brain parcellation
- **Research engineering:** Git, Linux/bash, high-performance computing (PBS), conda, reproducible workflows